TrustMind AI

Supervisor Briefing Notes | MSc Artificial Intelligence | UWE Bristol

Speakable project notes for supervisor meetings - From project start to current status

1. Opening - what the project is

TrustMind AI is an MSc Artificial Intelligence dissertation project at UWE Bristol.

I am building a text-based wellbeing check-in system that can assess early signs of distress from free text, with a strong focus on trust, reliability, and explainability - not diagnosis or therapy.

The live demo is at https://trustmind-ai.vercel.app, with the API hosted on Render.

2. Research question

My research question is:

To what extent does Retrieval-Augmented Generation (RAG) improve the trustworthiness, reliability, and explainability of LLM-generated wellbeing assessments compared with a standalone LLM?

In plain terms: if I take the same language model, does adding retrieval from trusted wellbeing sources make the outputs more reliable, more trustworthy, and easier to explain - versus the model answering from its own knowledge alone?

3. Why this problem matters

Standalone LLMs can sound confident but hallucinate or invent advice. In mental wellbeing, that is especially risky.

Commercial chatbots and analytics tools exist, but many are closed black boxes, or are not set up for a fair LLM versus LLM+RAG comparison with transparent methods.

The gap I am filling is an academic, controlled comparison of RAG for wellbeing assessment, using curated trusted sources, with a working demo to show it in practice.

4. Project journey - from the very start

Stage 1 - Dataset and EDA. I started with the SWMH dataset: Reddit posts labelled for depression, anxiety, SuicideWatch, bipolar, and offmychest. I ran exploratory analysis and preprocessing so the evaluation data was clean and consistent.

Stage 2 - LLM-only baseline. I built a standalone GPT-4.1 classifier with a fixed prompt and structured JSON output (label, confidence, reasoning), with no retrieval. I evaluated on 100 test posts, seed 42, temperature 0. Result so far: about 65% accuracy, macro-F1 around 0.65. That is my baseline.

Stage 3 - Knowledge base. Separately from SWMH, I built a curated knowledge base of trusted guidance - NHS, Student Minds, Samaritans, YoungMinds, UWE, and similar. Sources are manually approved URLs, not an open web crawl. Roughly 100+ approved URLs and about 85 cleaned documents.

Stage 4 - RAG pipeline. I chunked those documents, embedded them, and built hybrid retrieval: BM25 plus FAISS, fused with reciprocal rank fusion, top-5 passages into GPT-4.1. The model returns prediction, confidence, natural-language reasoning, and source IDs.

Stage 5 - Product demo. I shipped a full stack: Next.js frontend, FastAPI backend, with privacy controls, abstention if confidence is low, ethics disclaimers, and support links. Users can toggle LLM versus LLM+RAG on the same page for demos.

Stage 6 - Still to finish for the thesis numbers. I still need to run the full RAG evaluation on the same 100 samples and produce the side-by-side comparison notebook. The demo works; the formal RAG metrics file is the remaining experimental step.

5. What technology we are using - and why

A. Evaluation data - SWMH

- What: labelled Reddit wellbeing-related posts.

- Why: p

B. Knowledge collection - requests + BeautifulSoup + manual allow-list

- What: download only pre-approved pages; extract main text; store Markdown plus an audit CSV.

- Why: e

C. Embeddings - OpenAI text-embedding-3-small

- Why: good quality, cheap, simple, same vendor as the generator.

- Versus

D. Retrieval - hybrid BM25 + FAISS + RRF

- Why: wellbeing text needs both keyword matching (for example crisis terms) and semantic matching (paraphrases such as "worried about exams"). Hybrid covers both.

- Versus

E. Generator - GPT-4.1 (same in both arms)

- Why: strong instruction following and structured JSON; keeping the model fixed means any gain is from RAG, not from switching to a bigger model.

- Vers
question.

F. Application - Next.js + FastAPI + Vercel + Render

- Why: clear separation - the frontend never calls OpenAI; the backend owns the pipeline, keys, abstention, and logging.

- Versu
for LLM v

6. How this compares to what is in the market

Consumer wellbeing chatbots optimise for engagement and support chat, but are often closed and hard to measure for RAG lift.

Enterprise analytics RAG platforms optimise for business data Q&A - a different domain from wellbeing classification.

Pure fine-tuned classifiers can optimise label accuracy, but are weaker on explainability and trusted grounding.

Generic ChatGPT with no corpus is convenient, but has no controlled retrieval and a weak audit trail.

Our approach optimises for a fair LLM versus RAG comparison plus a transparent demo: curated sources, measurable metrics, and clear methods. We are not claiming to beat every commercial product. We are claiming a clear, defensible academic contribution: quantify what RAG adds for trustworthy wellbeing assessment.

7. How we measure "better"

We do not judge from one website click. We use:

- Reliability - accuracy, macro-F1, confusion matrix on the same 100 SWMH posts.

- Trustw

8. Current status (honest)

- LLM baseline: done (n=100, about 65% accuracy).

- Knowl

9. Closing line

TrustMind AI is a controlled study of whether hybrid RAG over a curated trusted wellbeing corpus improves reliability, trustworthiness, and explainability versus the same GPT-4.1 model alone - with a live student-facing demo to show the system in practice, and SWMH metrics to answer the research question rigorously.

Demo: https://trustmind-ai.vercel.app | API: https://trustmind-ai.onrender.com/health | Repo: github.com/Kushs22/trustmind-ai